



The Mediating Effects of Derived Relational Responding on the Relationship Between Verbal Operant Development and IQ

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Published online: 31 May 2018
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Abstract

Dixon, Belisle, and Stanley, (2018) demonstrated a strong, significant relationship between derived relational responding and intelligence in individuals with autism. We extended these results by evaluating the degree to which participant results on the PEAK Equivalence Pre-assessment (PEAK-E-PA, a direct assessment of derived relational responding) mediated the relationship between the PEAK Direct Training Assessment (PEAK-DT-A, a Skinnerian-based assessment of verbal development) and intelligence. Results support strong, positive correlations between both assessments (PEAK-DT-A and PEAK-E-PA) and IQ; however, the relationship between PEAK-DT-A and IQ could be explained in terms of participant results on the PEAK-E-PA alone. This finding corresponds with Relational Frame Theory, suggesting that derived responding can provide a behavioral interpretation of intelligent behavior, as well as Skinner's elementary operants.

Keywords Derived relational responding · RFT · Intelligence · Autism

General intelligence describes the several ways in which humans adapt to their environment (Spearman, 1904), and IQ scores are typically below average in individuals with autism and related disabilities. Although hypothetical mediating constructs such as “intelligence” are largely avoided in the behavioral sciences, several inductive behavioral theories have been advanced that describe the development of intelligent behavior, with implications for assessment and treatment for individuals with disabilities. One theory outlined in Skinner's *Verbal Behavior* (Skinner, 1957) describes the development of behaviors such as thinking, perceiving, and problem solving as the outcome of a history of reinforcement conferred upon a speaker by members of the speaker's verbal community. An emphasis in this theory is placed upon direct reinforcement for adaptive behavior. A second theory is described in *Relational Frame Theory* (RFT; Hayes, Barnes-Holmes, & Roche, 2001). According to RFT, intelligent behavior is the result of relatively few response classes, called relational operants, which describe ways that people respond to arbitrarily applicable events (e.g.,

words and symbols) relationally (i.e., in terms of other arbitrary and non-arbitrary events). For example, if a participant is directly reinforced for responding to a stimulus A in terms of both a stimulus B and a stimulus C, the participant may demonstrate the derived relational $B = C$ and $C = B$ response. This approach differs from Skinner's verbal behavior theory in several ways, but perhaps the greatest deviation is an emphasis on learning in the absence of direct acting contingencies (Healy, Barnes-Holmes, & Smeets, 2000). Behaviors conventionally measured on IQ tests are unlikely to have been directly reinforced (e.g., block design, analogical reasoning), therefore deriving relationships between test items and prior conditional discriminations may be necessary for strong performance. Because both Skinner's verbal operant theory and RFT provide behavioral accounts of language and cognitive development, convergent validity with intelligence test outcomes may be expected.

Two studies published to date have evaluated the relationship between these theoretical approaches and IQ with individuals with autism and related disabilities, using assessments contained in the PEAK Relational Training System (PEAK; Dixon, 2014, 2015). PEAK is an assessment and curriculum package designed to promote language and cognitive skills using verbal behavior approaches along with advances in RFT, in application with individuals with disabilities. Dixon, Whiting, Rowsey, and Belisle, (2014) compared participant scores on the PEAK Direct Training Assessment (PEAK-

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DT-A; Dixon, 2014) and IQ assessment scores accessed from existing client records. The PEAK-DT-A is a criterion-referenced assessment that combines direct and indirect assessment methods to provide a measure of participants' verbal development consistent with a Skinnerian verbal operant approach (Dixon, Whiting, & Daar, 2014). Items on the assessment range from simple (e.g., echoics, tacts) to complex (e.g., metonymical tacts, audience control) forms of verbal behavior described by Skinner, (1957). Results indicated a strong, positive correlation between outcomes on the PEAK-DT-A and participant scores on several IQ assessments. In the second study, Dixon et al., (2018) evaluated the relationship between participant scores on the PEAK Equivalence Pre-assessment (PEAK-E-PA; Dixon, 2015) and IQ. The PEAK-E-PA is another criterion-referenced assessment that is implemented directly with participants and provides a measure of arbitrarily applicable derived relational responding consistent with RFT (Dixon et al., 2018). Items on the PEAK-E-PA assess the degree to which participants derive reflexive, symmetrical, transitive, and equivalence relations (see Sidman & Tailby, 1982 for a review; note, that although stimuli can be related in ways other than by sameness, only relations contained in an equivalence account are tested in the PEAK-E-PA).

The purpose of the present study was to extend results reported by Dixon et al., (2018) by determining whether participant results on the PEAK-E-PA served as a mediating variable for a relationship between results on the PEAK-DT-A and IQ. A *mediating variable* is a third variable that accounts for the relationship between a predictor variable and the predicted outcome. Derived relational responding is not only theoretically central to the interpreting intelligent behavior (Hayes et al., 2001), but may also explain the development of the verbal operant response classes described by Skinner. If adding participant PEAK-DT-A scores to the model reported by Dixon, Belisle, and Stanley improved prediction of IQ, this would suggest that assessments of both derived relational responding as well as verbal operant development should be considered in a behavioral account of intelligent behavior. Conversely, if adding PEAK-DT-A provided no statistical benefit, this would suggest that derived relational responding mediates the relationship between verbal operant development and IQ, supporting Hayes et al., (2001) position that intelligence is likely the outcome of relatively few relational operants, with potentially large implications for assessment and treatment for individuals with disabilities.

Methods

Participants and Setting

Participants were 55 males and nine females between the ages of 5–16 years ($M = 11.43$, $SD = 3.6$). Of the 64 participants,

59 had a diagnosis of autism (92%), and the others had a diagnosed disability other than autism (8%). All participants attended a Midwestern school specialized for individuals with autism. Sixty of the participants in the present study had also previously participated in Dixon et al., (2018), and their IQ and PEAK-E-PA scores from that study were also used as the scores reported in the current study (permission from the authors was obtained prior to this study). IQ scores ranged from 48 to 132, suggesting that participants were representative of varying degrees of intellectual functioning.

All assessments were completed either in the participants' home rooms or in a separate room. All classrooms contained between three and eight students, one or two classroom teachers, and two or three paraprofessionals. When the assessments were conducted in the classrooms, they were conducted in an area away from the students, teachers, and paraprofessionals to minimize distractions. The additional rooms in which some assessments took place minimally contained a table, two chairs, and preferred items.

Materials and Procedure

The study was conducted across an 8-month period, during which participants were assessed using the PEAK-DT-A, PEAK-E-PA, and WISC-IV. All three assessments for a given participant were completed within a 6-month period, as in the procedures by Dixon et al., (2018). Because PEAK-E-PA and IQ scores were obtained from a prior study, PEAK-DT-A scores were always the last to be obtained. The assessments and materials used are discussed in the following subsections.

PEAK-Direct Training Assessment (PEAK-DT-A) PEAK-DT-A is a 184-item criterion-referenced assessment that identifies verbal operant skills that participants can and cannot perform, ranging from simple forms of verbal operant behavior (e.g., tacts and mands) to more complex forms of verbal operant behavior (e.g., metonymical tacts and audience control). The test is completed using a combination of direct and indirect assessment measures, and a combination of both methods has been reported in prior literature (e.g., Dixon, Whiting, Rowsey, & Belisle, 2014). Each item on the assessment has three "checkboxes" where assessors indicate the participant has, or has not, been observed performing the skill in the past, or they are unsure whether the participant has performed the skill in the past. Responses can be directly tested using a standard discrete trial testing arrangement, which is recommended when item assessors were uncertain. In the present study, assessments were first completed indirectly by trained graduate students who had at least 4 months of experience working directly with the participants. The graduate students were blind to the participants' PEAK-E-PA and IQ assessment scores while completing the assessment. Then, when assessors indicated that they were unsure whether the participant had

demonstrated the skill, the skill was directly tested in a 10-trial block. Ten-trial blocks were used based on recommendations made by Dixon, (2014), as well as to reduce the probability of correct responding occurring by chance alone. For each trial, the assessor presented the participant with a discriminative stimulus and allowed 3–5 s for the participant to provide a response. If nine correct responses were provided within the block (90%), the item was considered mastered. The total number of items was then summed for a PEAK-DT-A total score ranging from 0 to 184. Interobserver agreement was completed for 31.25% of the assessments by having a second graduate student familiar with the participants complete the indirect assessment. The second assessor was blind to the results provided by the initial assessor. For all items that were scored as either “Yes” or “No,” the percentage of agreement between the two observers was evaluated, producing IOA of 92% (range, 84 to 100%).

PEAK-Equivalence Pre-assessment (PEAK-E-PA) PEAK-E-PA is a 48-item criterion-referenced assessment that assesses the degree to which a participant’s responding is under the control of reflexive, symmetrical, transitive, and equivalence relations. The assessment is implemented directly, with 12 items for each type of derived relation. For reflexivity items, the participant is provided a sample stimulus (A) and an identical stimulus (A), and is told by the assessor that these items are the same. The participant is then provided the sample stimulus (A) and an array of comparison stimuli; a correct response requires the selection of the identical (A) stimulus from the array. For symmetry items, the participant is told that a sample stimulus (A) is the same as a comparison stimulus (B), and is required to select the stimulus (A) from an array of comparison stimuli when presented the stimulus (B) as the sample stimulus. For transitivity and equivalence items, the participant is told two relations by the assessor and a third relation is tested (e.g., directly train $A = B$ and $B = C$, and test either $A = C$ or $C = A$). Stimulus relations assessed are arbitrary and include cross-modal stimulus relations (e.g., olfactory and gustatory). In the current study, the assessments were again conducted by trained graduate students. The number of total correct responses was summed to arrive at a PEAK-E-PA total score ranging from 0 to 48. IOA was evaluated by having a second observer record participant responses on 23.4% of the assessments. IOA was 100%.

WISC-IV Intelligence Assessment Short Form (WISC-IV) The Short Form version of the WISC-IV assessment was used in the current study, which included the Vocabulary and Block Design subtests. As noted by Sattler and Dumont, (2004), the combined scores from these two subtests provide a measure of full scale IQ that strongly correlates with the full assessment; therefore, the subtests’ scores provide a valid measure of intellectual functioning for research. For the block design

subtest, participants are provided with a model block design and an assortment of half-red-half-white blocks. The participant is then given an amount of time to build a block model with point-to-point correspondence with the exemplar model. Block models increase in complexity throughout the assessment. For the vocabulary subtest, participants are either given a picture and are required to name the picture, or are given a vocal word and are required to describe the vocal word. Pictures and vocal words increase in complexity throughout the assessment. In the current study, both subtests were completed by trained graduate students directly with the participants. The scores from both subtests were summed to provide a total raw IQ score and scaled to provide a normative full-scale IQ score. Both raw and full-scale IQ scores were used in all analyses in the current study. IOA was evaluated by having a second observer record participant responses on 23.4% of the assessments. IOA was 100%.

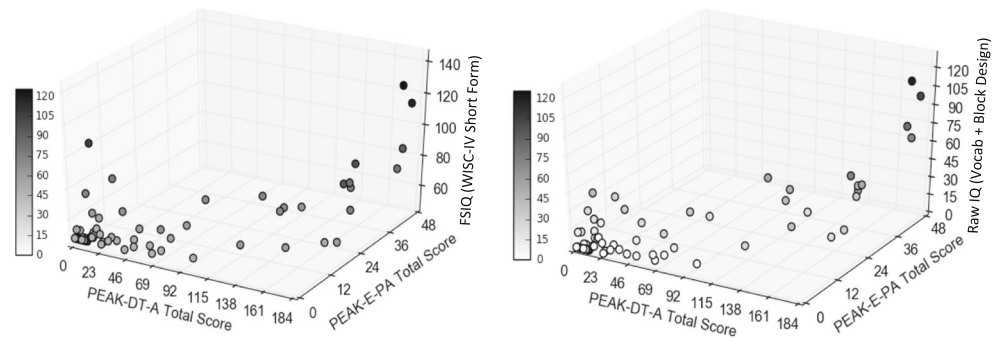
Statistical Analyses

Two statistical analyses were conducted in the current study. First, Pearson correlational analyses were conducted to evaluate the relationship between PEAK-DT-A total scores and IQ and between PEAK-E-PA total scores and IQ. Second, a hierarchical multiple linear regression analysis was conducted to determine if either the PEAK-DT-A or the PEAK-E-PA served as a mediating variable between either of the relationships identified in the prior correlational analyses. The regression model evaluates mediating variables by comparing predictive models that contain only a single component (e.g., PEAK-E-PA alone) to predictive models containing both components (e.g., $PEAK-E-PA * PEAK-DT-A$), in terms of a given outcome variable (IQ), to determine if additional components contribute to the predictive power of a given model.

Results and Discussion

The results of the study are summarized in Fig. 1 and Table 1. The left panel of Fig. 1 shows a three-dimensional scatterplot of participant scores on PEAK-DT-A (variable, X1) and the PEAK-E-PA (variable, X2) as predictors of full scale IQ (variable, Y). The right panel displays the same information, presenting raw IQ values in place of full scale IQ. Visual analysis of both graphs shows that low scores on either PEAK-DT-A or PEAK-E-PA were predictive of low IQ scores, and high scores on either assessment were indicative of high IQ scores. A Pearson’s correlational analysis confirms these results, suggesting a strong, positive correlation between both the PEAK-DT-A and IQ (full scale IQ $r = 0.66$, $p < 0.01$; raw IQ 0.80 , $p < 0.01$) and the PEAK-E-PA and IQ (full scale IQ 0.85 , $p < 0.01$; raw IQ 0.89 , $p < 0.01$). Table 1 provides the results from the hierarchical multiple linear regression analysis,

Fig. 1 Three-dimensional scatterplot of participant scores on the PEAK-DT-A (X1) and the PEAK-E-PA (X2) as predictors of full scale IQ (Y; left) and raw IQ (Y; right). Individual data points are plotted and shaded based on participant's obtained IQ scores



comparing the predictive utility of the combined PEAK-DT-A * PEAK-E-PA model to both the PEAK-DT-A and PEAK-E-A models in isolation, in terms of full scale IQ and raw IQ. The first set of tested models used full scale IQ as the outcome variable. The first model, which used PEAK-DT-A only, was a predictor of participant IQ scores, and adding PEAK-E-PA scores provided significantly greater predictive utility to the model. The second model, which used PEAK-E-A only, was also a significant predictor of IQ scores; however, adding PEAK-DT-A scores to the model failed to provide any greater predictive utility. The same model comparisons were evaluated using raw IQ and produced the same outcome, suggesting that PEAK-DT-A scores provided no greater utility in predicting IQ when added to a model consisting exclusively of participant scores on the PEAK-E-PA.

These results support the position that the positive correlation between participant scores on PEAK-DT-A and IQ may be accounted for by the degree to which participants' respond in accordance with derived relations (Hayes et al., 2001),¹ consistent with an RFT account of language development. Our results replicate findings reported by suggesting a strong, significant correlation between PEAK-DT-A and IQ. In addition, we report a stronger, significant correlation between PEAK-E-PA and IQ than the results of Dixon, Whiting, and Daar, (2014). The combined model (PEAK-DT-A * PEAK-E-PA), however, failed to produce greater predictive power in terms of IQ compared to the use of participant scores on the PEAK-E-PA alone. This outcome suggests that adding PEAK-DT scores as a measure of verbal operant development is unlikely to provide more predictive utility than simply evaluating derived relational responding, in terms of predicting intellectual functioning. A first implication of these results is theoretical, in that the data support the general position put forward by Hayes et al., (2001) that intelligence may be accounted for by relatively few, arbitrarily applicable relational operants. A second implication is practical and corresponds with the first; if intelligence is simply the result of arbitrary relational responding, then improving intelligence scores would be achieved through the use of instructional strategies

that promote the emergence of relational responding. This implication exceeds the current correlational analysis, but two other studies have been published recently that support this general conclusion. Dixon, Belisle, Blevins, and Hayes, (under review) recently provided the first evidence that training multiple exemplars of socially relevant derived stimulus relations can lead to improvements in derived relational responding across novel exemplars in children with autism. Second, Cassidy, Roche, and Hayes, (2011) demonstrated that relational training conducted with typically developing children can lead to large increases in IQ. Though these studies are promising, more research is needed in this area to better determine how behavioral instructional strategies that promote the emergence of relational responding can lead to improvements in IQ scores.

A first and potentially major limitation was that the PEAK-DT-A was always conducted as the final assessment, with up to a 6-month delay between assessment periods. Although prior research has suggested that scores on this assessment are unlikely to change significantly within 1-month (Dixon, Stanley, Belisle, & Rowsey, 2016), this limitation does decrease certainty in the multiple regression comparison. If drift had occurred across assessments, however, lower correlations

Table 1 Multiple hierarchical linear regression analysis of PEAK-DT-A and PEAK-E-A as predictors of IQ (WPPSI-III; WISC-IV)

Model	<i>R</i>	<i>R</i> ²	Adjusted <i>R</i> ²	<i>F</i> change
FSIQ (DT, E)				
1: DT only	.656	.430	.421	46.8**
2: DT * E	.805	.648	.637	37.8**
FSIQ (E, DT)				
1: E only	.803	.646	.640	112.9**
2: E * DT	.805	.648	.637	0.4
Raw IQ (DT, E)				
1: DT only	.796	.634	.628	107.5**
2: DT * E	.891	.794	.788	47.5**
Raw IQ (E, DT)				
1: E only	.888	.788	.784	230.2**
2: E * DT	.891	.794	.788	1.9

p* < .05; *p* < .01

¹ Only equivalence relations were assessed in the current study, where several other forms of relating are possible (Hayes, Barnes-Holmes, & Roche).

would be expected between PEAK-DT-A and the other measures, so the reported relationships may be stronger than what the present study indicates. In an ideal design, the order of assessments would have been randomized, and the time between assessments minimized to within 1-month given prior test-retest results. Second, participants represented a convenience sample from the same location. Research on derived relational responding and verbal operant development with individuals with autism is both interesting and needed given language deficits central to this disorder, and larger scale research is needed to better understand how language develops from a behavior analytic perspective. Third, differences in how the PEAK-DT-A and PEAK-E-PA are conducted limit the scope of the obtained results, such that a clear comparison of Skinnerian-type operant conditioning and strictly relational learning was not achieved in the current analysis. The PEAK-DT-A was developed from Skinnerian verbal behavior theories, but uses both indirect and direct assessment methods to determine participants' skill repertoires; whereas the PEAK-E-PA provides a direct assessment of derived relational responding using a train and test strategy. Therefore, differences in the two measures in terms of their relationship to IQ could be an artifact of the different ways in which these assessments are conducted. An alternative may be to directly test all items on the PEAK-DT-A, but testing in 10-trial blocks with an 80 to 90% independent correct responding mastery criterion as suggested by Dixon, Whiting, and Daar, (2014) would require 1840 testing trials, providing greater precision at the expense of any meaningful real-world application. This limitation could be addressed in future research using the recently developed PEAK-Direct Training Pre-assessment (PEAK-DT-PA; PEAKaba.com, 2017), which provides a sample of 64 items from the PEAK-DT-A that are directly tested with participants, like the PEAK-E-PA. More research is needed on the PEAK-DT-PA before it can be used in more extensive evaluations such as is provided in the current investigation. Finally, IQ scores obtained for the purposes of the current investigation were derived from the Short Form version of the WISC-IV. As noted by Sattler and Dumont, correlations between scores obtained on the Short Form version and full test administration scores support the use of this method in research; however, this limitation could be addressed in future research to obtain more information on how derived relational responding relates to the several adaptive skills that are evaluated in traditional IQ assessments. Beyond addressing the above limitations, future research in this area may compare measures of derived relational responding to other Skinnerian-based language assessments, such as the *Verbal Behavior Milestones Assessment and Placement Program* (VB-MAPP; Sundberg, 2008), and their relationship to the development of intelligent behavior. Future research may also evaluate how deriving relations other than sameness (e.g.,

distinction, relating hierarchically) differentially accounts for intellectual development.

In summary, prior research has supported both the relationship between the PEAK-DT-A as a Skinnerian-based assessment of language development and the PEAK-E-PA as a measure of derived relational responding, as predictors of IQ in individuals with autism (Dixon, Whiting, Rowsey, & Belisle, 2014; Dixon et al., 2018). The current study extends upon these findings suggesting that the relationship between PEAK-DT-A and IQ may be accounted for by participants' abilities to derive relations as measured on the PEAK-E-PA. This finding is consistent with RFT, with implications for understanding how language develops from a behavior analytic perspective, and for the assessment and treatment of language deficits experienced by individuals with autism and related disabilities.

Compliance with Ethical Standards

Informed Consent Informed consent was obtained from the participants, agencies, and/or legal guardians of all participants included in this study.

Ethical Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Conflict of Interest The second author receives small royalties from sales of the PEAK curriculum. The other authors declare that they have no conflict of interest.

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